SCIENCE 4
TERM 1 WEEK 3
DAY 1

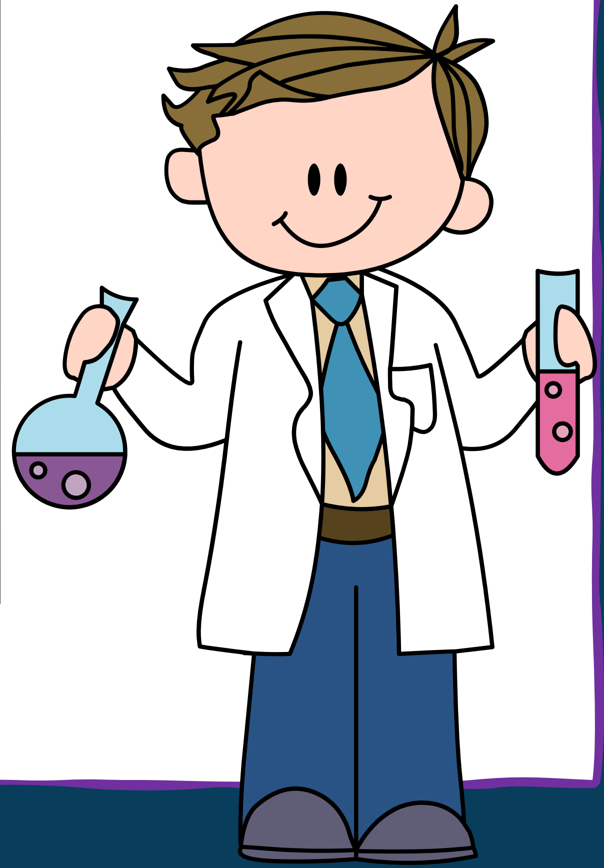
PRE-LESSON

Directions: Give five (5) science inventions.

1.
2.
3.
4.
5.

PRE-LESSON

How can they compare what the children did to the paper with what the teacher did?



CHEMICAL PROPERTIES OF MATERIALS

CHEMICAL PROPERTIES

Chemical properties of materials are characteristics that determine how a substance interacts with other substances during chemical reactions, including its reactivity, flammability, and ability to oxidize.

CHEMICAL PROPERTIES

These properties can only be observed when a material undergoes a chemical change, resulting in the formation of new substances.

COMBUSTIBILITY

Some materials can be burnt, meaning they can undergo combustion reactions in the presence of oxygen, producing heat and often releasing gases and ash. Examples include wood, paper, and certain plastics.

COMBUSTIBILITY

Example: Wood is combustible because it can be burned to produce heat and ash.

Specific Example: Paper, a type of combustible material, burns readily when exposed to flame or heat.

FLAMMABILITY

Materials vary in their ability to ignite and burn in the presence of a flame or heat source. **Flammability** is a critical property to consider for safety and regulatory purposes, influencing their use in construction, manufacturing, and transportation.

FLAMMABILITY

Example: Cotton fabric is flammable and can catch fire easily when exposed to a flame.

Specific Example: Polyethylene, a type of plastic, exhibits high flammability and can ignite quickly when exposed to a spark or flame.

REACTIVITY

Materials can react with other substances, leading to chemical changes. For instance, metals can react with acids to produce salts and hydrogen gas. Chemical reactivity varies widely among materials based on their composition and structure.

REACTIVITY

Example: Metals exhibit reactivity when they react with acids to produce salts and release hydrogen gas.

REACTIVITY

Specific Example: Iron undergoes a chemical reaction with oxygen and moisture in the air to form rust (iron oxide), which weakens the metal over time.

DEGRADABILITY

This refers to the ability of materials to break down physically or chemically over time. Degradable materials can be broken down into smaller components through processes such as weathering, mechanical action, or microbial activity.

DEGRADABILITY

Example: Leaves on the ground degrade naturally through exposure to weather and microbial activity.

Specific Example: Biodegradable plastics degrade into simpler substances when exposed to sunlight and water, unlike conventional plastics that persist for much longer.

BIODEGRADABILITY

Specifically, biodegradable materials can be broken down by biological organisms, such as bacteria and fungi, into simpler compounds like water, carbon dioxide, and biomass. Examples include natural fibers, certain plastics, and organic matter.

BIODEGRADABILITY

Example: Organic matter like food scraps can biodegrade when exposed to microbial activity in compost bins.

Specific Example: Biodegradable packaging made from cornstarch can be composted, breaking down into natural components without leaving harmful residues in the environment.

ASSESSMENT

Directions: Give the chemical properties of each material.

1. 

ASSESSMENT

2. 

3. 

ASSESSMENT

4.

5.

EXTENDED

Directions: Fill in the table with examples of the chemical properties of matter.

CHEMICAL PROPERTIES	EXAMPLES
1. Combustibility	1. 2.

EXTENDED

CHEMICAL PROPERTIES	EXAMPLES
2. Flammability	1. 2.
3. Reactivity	1. 2.

EXTENDED

CHEMICAL PROPERTIES	EXAMPLES
4. Degradability	1. 2.
5. Biodegradability	1. 2.

SCIENCE 4

TERM 1 WEEK 3

DAY 2

PRE-LESSON

Directions: Write **CHEMICAL** if the statement is correct, **PHYSICAL** if not.

________ 1. Wood, which is combustible, can be burned to produce heat and ash.

PRE-LESSON

2. Propane, used in gas grills, is flammable and ignites easily when mixed with air.

3. Baking soda reacts with vinegar to produce carbon dioxide gas and a bubbling effect.

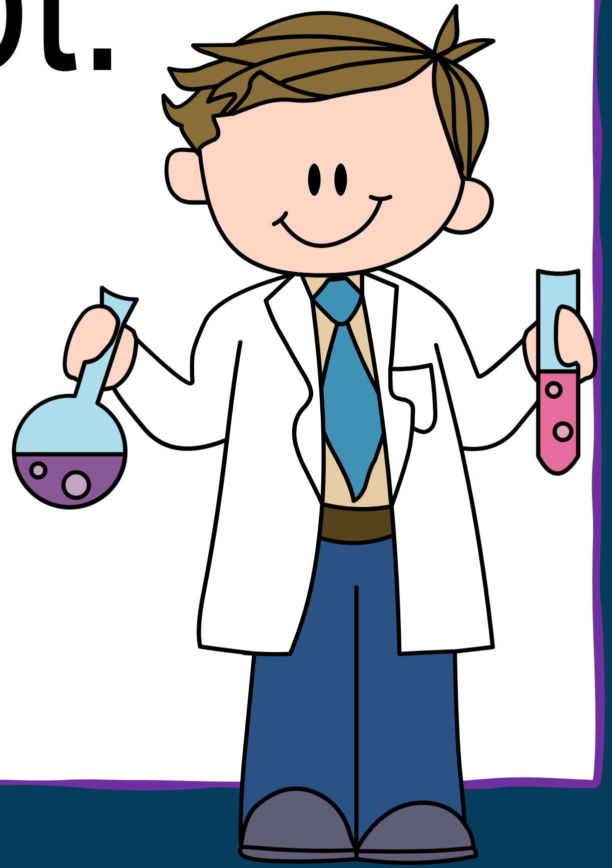
PRE-LESSON

________ 4. Leaves on the ground naturally degrade through exposure to weather and microbial activity.

________ 5. Leaves and other plant materials biodegrade in a compost pile, turning into rich soil.

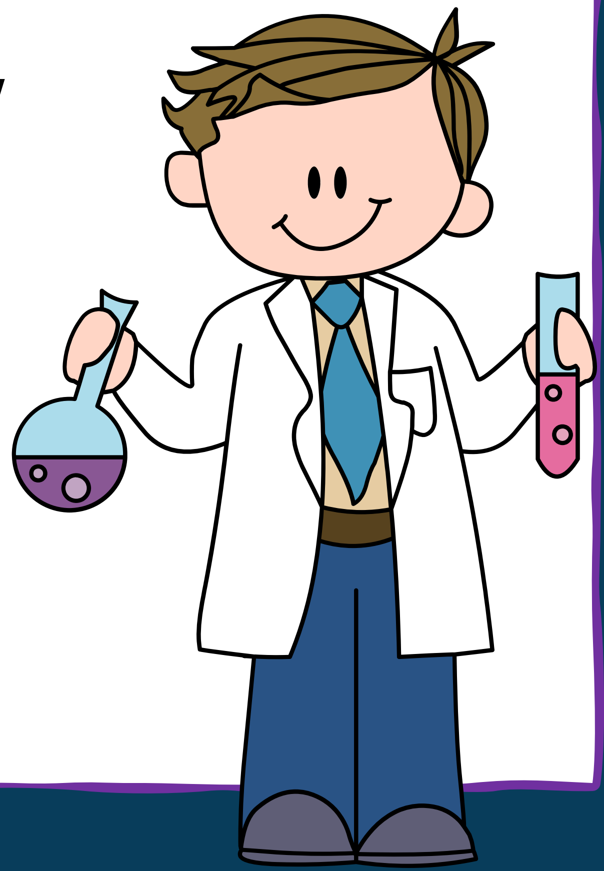
PRE-LESSON

Directions: Draw 😊 if the picture corresponds to the chemical property, ☹ if it does not.



1.

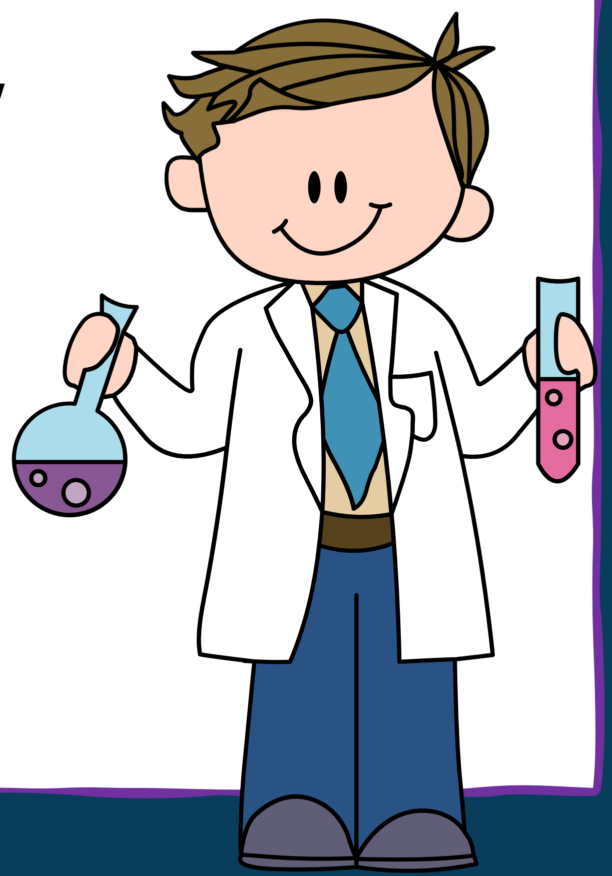
Flammability



PRE-LESSON

2.

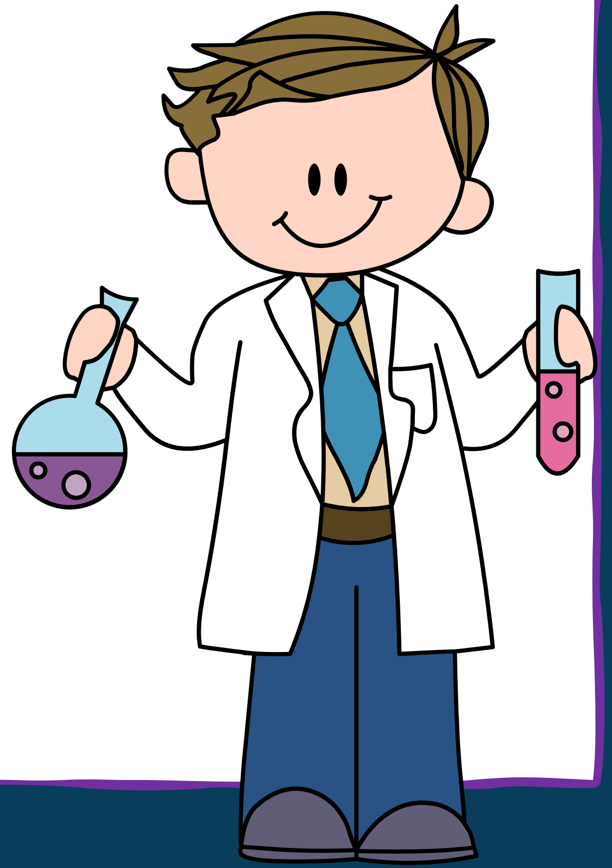
Flammability



PRE-LESSON

3.

Reactivity



PRE-LESSON

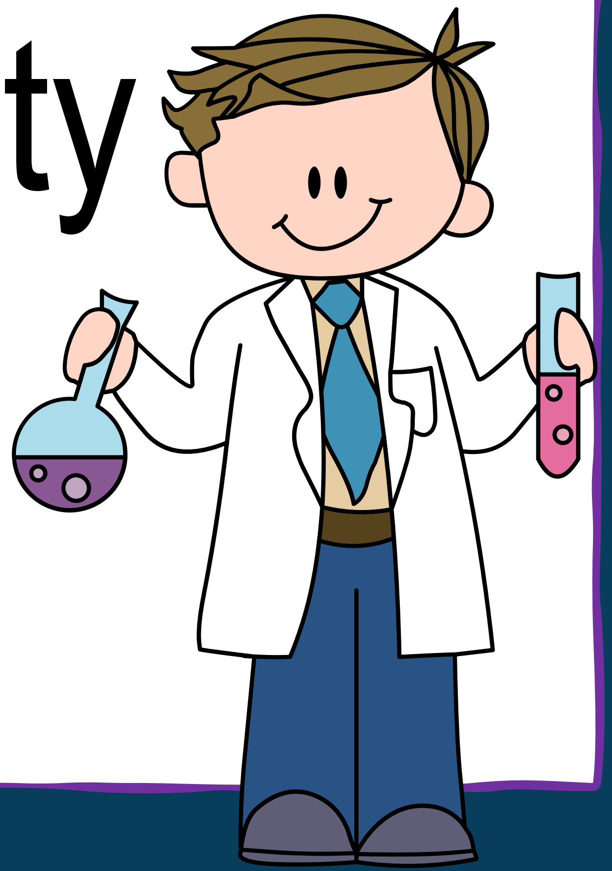
4.

Degradability

PRE-LESSON

5.

Biodegradability



CHEMICAL PROPERTIES OF MATERIALS

CHEMICAL PROPERTIES

Chemical properties of materials are characteristics that determine how a substance interacts with other substances during chemical reactions, including its reactivity, flammability, and ability to oxidize.

CHEMICAL PROPERTIES

These properties can only be observed when a material undergoes a chemical change, resulting in the formation of new substances.

COMBUSTIBILITY

Some materials can be burnt, meaning they can undergo combustion reactions in the presence of oxygen, producing heat and often releasing gases and ash. Examples include wood, paper, and certain plastics.

COMBUSTIBILITY

Example: Wood is combustible because it can be burned to produce heat and ash.

Specific Example: Paper, a type of combustible material, burns readily when exposed to flame or heat.

FLAMMABILITY

Materials vary in their ability to ignite and burn in the presence of a flame or heat source. **Flammability** is a critical property to consider for safety and regulatory purposes, influencing their use in construction, manufacturing, and transportation.

FLAMMABILITY

Example: Cotton fabric is flammable and can catch fire easily when exposed to a flame.

Specific Example: Polyethylene, a type of plastic, exhibits high flammability and can ignite quickly when exposed to a spark or flame.

REACTIVITY

Materials can react with other substances, leading to chemical changes. For instance, metals can react with acids to produce salts and hydrogen gas. Chemical reactivity varies widely among materials based on their composition and structure.

REACTIVITY

Example: Metals exhibit reactivity when they react with acids to produce salts and release hydrogen gas.

REACTIVITY

Specific Example: Iron undergoes a chemical reaction with oxygen and moisture in the air to form rust (iron oxide), which weakens the metal over time.

DEGRADABILITY

This refers to the ability of materials to break down physically or chemically over time. Degradable materials can be broken down into smaller components through processes such as weathering, mechanical action, or microbial activity.

DEGRADABILITY

Example: Leaves on the ground degrade naturally through exposure to weather and microbial activity.

Specific Example: Biodegradable plastics degrade into simpler substances when exposed to sunlight and water, unlike conventional plastics that persist for much longer.

BIODEGRADABILITY

Specifically, biodegradable materials can be broken down by biological organisms, such as bacteria and fungi, into simpler compounds like water, carbon dioxide, and biomass. Examples include natural fibers, certain plastics, and organic matter.

BIODEGRADABILITY

Example: Organic matter like food scraps can biodegrade when exposed to microbial activity in compost bins.

Specific Example: Biodegradable packaging made from cornstarch can be composted, breaking down into natural components without leaving harmful residues in the environment.

ASSESSMENT

EXPERIMENT TIME!

ASSESSMENT

BOTTLE BALLOON

Materials Needed:

- Baking soda (sodium bicarbonate)
- Vinegar (acetic acid)
- A small container or a cup
- A balloon (optional, to capture the gas)
- A spoon

ASSESSMENT

Procedure:

1. Preparation:

- Gather all materials.
- Measure about 1-2 tablespoons of baking soda.
- Measure about 1/4 cup of vinegar.

ASSESSMENT

2. Experiment Setup:

- Place the baking soda into the small container or cup.
- If you are using a balloon, stretch the opening of the balloon over the top of the container, making sure it's securely in place.

ASSESSMENT

3. Conducting the Experiment:

- Pour the vinegar into the container with the baking soda. If using a balloon, make sure the balloon is properly attached to capture the gas produced.

- Observe the reaction.

ASSESSMENT

3. Conducting the Experiment:

- Observe the reaction. You should see bubbling and fizzing as the vinegar reacts with the baking soda. If you are using a balloon, it will start to inflate.

ASSESSMENT

3. Conducting the Experiment:

- Observe the reaction. You should see bubbling and fizzing as the vinegar reacts with the baking soda. If you are using a balloon, it will start to inflate.

EXTENDED

Directions: From the experiment, answer the following questions:

1. What observable signs indicate that a chemical reaction is occurring between the baking soda and vinegar?

EXTENDED

2. What are the products formed when baking soda reacts with vinegar?

EXTENDED

3. Why might a balloon be used in this experiment, and what does its inflation signify?

EXTENDED

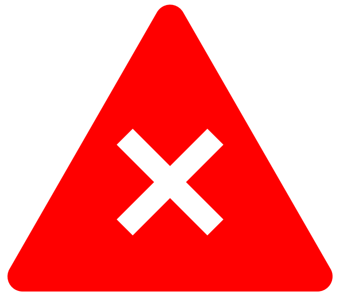
4. Which chemical property of baking soda is demonstrated by this experiment?

EXTENDED

5. CONCLUSION:

SCIENCE 4
TERM 1 WEEK 3
DAY 3

PRE-LESSON

Directions: Draw  if the statement is correct,  if not.

______ 1. Cotton fabric, known for its flammability, can catch fire easily when exposed to a flame.

PRE-LESSON

2. Iron undergoes a chemical reaction with oxygen and moisture in the air to form rust (iron oxide), which weakens the metal over time.

PRE-LESSON

3. Cardboard boxes degrade over time when exposed to moisture and environmental conditions.

PRE-LESSON

4.Organic matter like food scraps can biodegrade when exposed to microbial activity in compost bins.

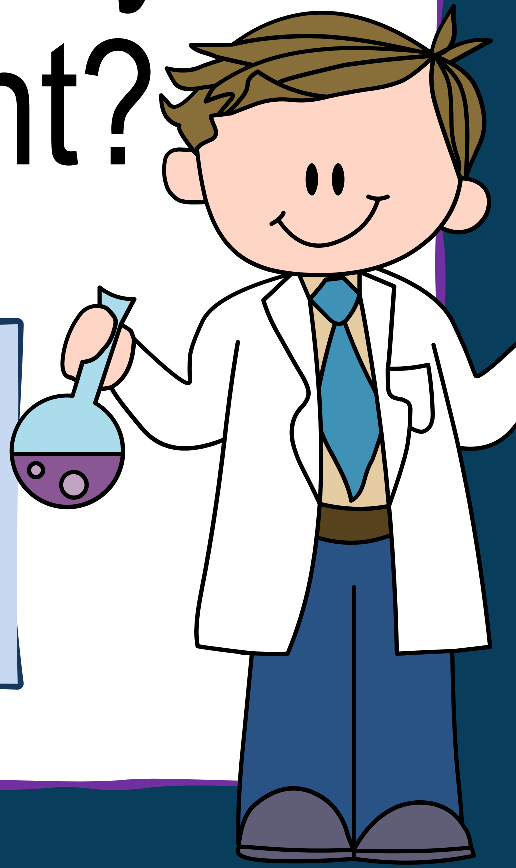
5. Coal is a combustible material that produces energy when burned.

PRE-LESSON

1. How can you show empathy and responsibility by reducing the use of non-biodegradable materials in your daily life?

PRE-LESSON

2. List three ways you can help reduce the amount of non-biodegradable materials used at home or school. How would these actions demonstrate empathy and responsibility towards the environment?



CHEMICAL PROPERTIES OF MATERIALS

CHEMICAL PROPERTIES

Chemical properties of materials are characteristics that determine how a substance interacts with other substances during chemical reactions, including its reactivity, flammability, and ability to oxidize.

CHEMICAL PROPERTIES

These properties can only be observed when a material undergoes a chemical change, resulting in the formation of new substances.

COMBUSTIBILITY

Some materials can be burnt, meaning they can undergo combustion reactions in the presence of oxygen, producing heat and often releasing gases and ash. Examples include wood, paper, and certain plastics.

COMBUSTIBILITY

Example: Wood is combustible because it can be burned to produce heat and ash.

Specific Example: Paper, a type of combustible material, burns readily when exposed to flame or heat.

FLAMMABILITY

Materials vary in their ability to ignite and burn in the presence of a flame or heat source. **Flammability** is a critical property to consider for safety and regulatory purposes, influencing their use in construction, manufacturing, and transportation.

FLAMMABILITY

Example: Cotton fabric is flammable and can catch fire easily when exposed to a flame.

Specific Example: Polyethylene, a type of plastic, exhibits high flammability and can ignite quickly when exposed to a spark or flame.

REACTIVITY

Materials can react with other substances, leading to chemical changes. For instance, metals can react with acids to produce salts and hydrogen gas. Chemical reactivity varies widely among materials based on their composition and structure.

REACTIVITY

Example: Metals exhibit reactivity when they react with acids to produce salts and release hydrogen gas.

REACTIVITY

Specific Example: Iron undergoes a chemical reaction with oxygen and moisture in the air to form rust (iron oxide), which weakens the metal over time.

DEGRADABILITY

This refers to the ability of materials to break down physically or chemically over time. Degradable materials can be broken down into smaller components through processes such as weathering, mechanical action, or microbial activity.

DEGRADABILITY

Example: Leaves on the ground degrade naturally through exposure to weather and microbial activity.

Specific Example: Biodegradable plastics degrade into simpler substances when exposed to sunlight and water, unlike conventional plastics that persist for much longer.

BIODEGRADABILITY

Specifically, biodegradable materials can be broken down by biological organisms, such as bacteria and fungi, into simpler compounds like water, carbon dioxide, and biomass. Examples include natural fibers, certain plastics, and organic matter.

BIODEGRADABILITY

Example: Organic matter like food scraps can biodegrade when exposed to microbial activity in compost bins.

Specific Example: Biodegradable packaging made from cornstarch can be composted, breaking down into natural components without leaving harmful residues in the environment.

ASSESSMENT

Directions: Answer the following questions.

1. How is burning paper similar to burning wood, and why should we be cautious when they are near flames?

ASSESSMENT

2. How does cotton fabric catching fire quickly compare to plastic toys when exposed to heat? Why is it important to understand this difference?

ASSESSMENT

3. How is the process of iron rusting similar to how food scraps break down in a compost bin when they get wet? What happens to them as time goes by?

ASSESSMENT

4. How do leaves decomposing on the ground over time resemble the special plastic that breaks down in sunlight and water? What distinguishes them from regular plastic?

ASSESSMENT

5. How is using biodegradable packaging made from cornstarch similar to utilizing natural materials like wood and paper? Why is this beneficial for the environment?

EXTENDED

Directions: Place the following materials with their corresponding chemical properties in the table.

wood coal dry leaves gasoline
alcohol rusty window food waste
paper cardboard ecobag

EXTENDED

Combustibility	Flammability	degradability	Biodegradability	Reactivity

SCIENCE 4
TERM 1 WEEK 3
DAY 4

PRE-LESSON

Directions: Answer the following questions.

- **Flammability and Combustibility:**

1. How are flammability and combustibility similar?

2. What happens when something is flammable or combustible?

PRE-LESSON

3. Can you give an example of something that is flammable or combustible?

PRE-LESSON

- **Biodegradability and Degradability:**

1. How are biodegradability and degradability similar?

2. What does it mean when something is biodegradable or degradable?

PRE-LESSON

3. Name something around you that is biodegradable or degradable.

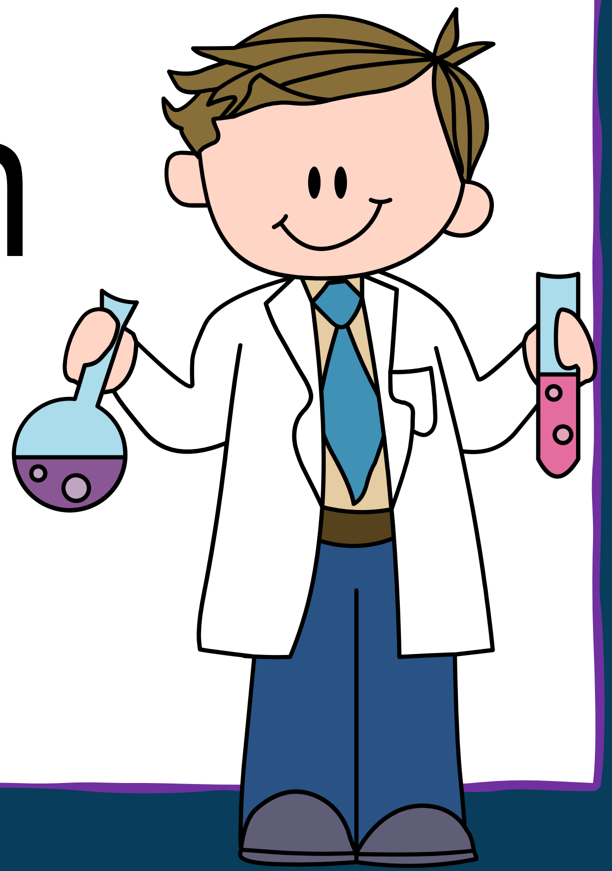
PRE-LESSON

Describe the chemical properties of the following materials.

1. Burning woods

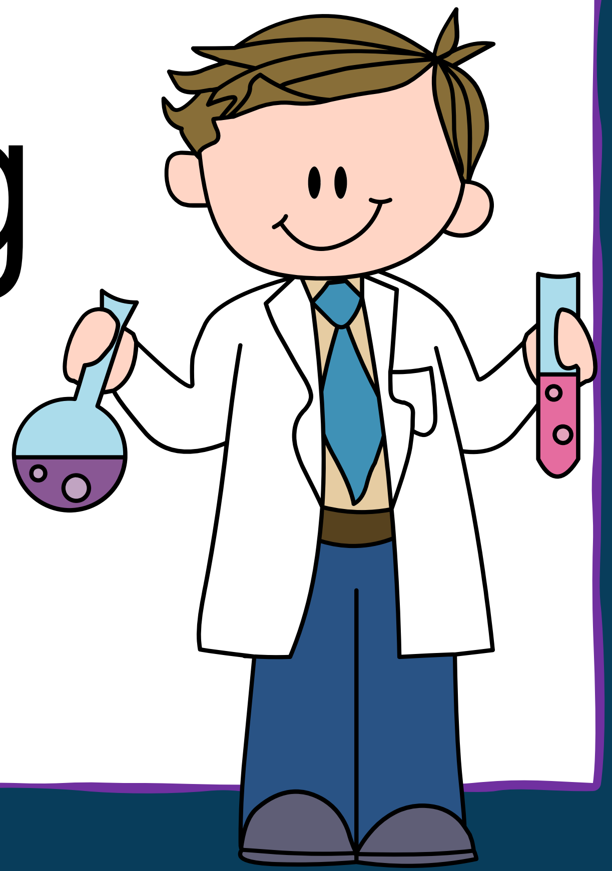
PRE-LESSON

2. Combustion of Gasoline



PRE-LESSON

3. Composting



CHEMICAL PROPERTIES OF MATERIALS

CHEMICAL PROPERTIES

Chemical properties of materials are characteristics that determine how a substance interacts with other substances during chemical reactions, including its reactivity, flammability, and ability to oxidize.

CHEMICAL PROPERTIES

These properties can only be observed when a material undergoes a chemical change, resulting in the formation of new substances.

COMBUSTIBILITY

Some materials can be burnt, meaning they can undergo combustion reactions in the presence of oxygen, producing heat and often releasing gases and ash. Examples include wood, paper, and certain plastics.

COMBUSTIBILITY

Example: Wood is combustible because it can be burned to produce heat and ash.

Specific Example: Paper, a type of combustible material, burns readily when exposed to flame or heat.

FLAMMABILITY

Materials vary in their ability to ignite and burn in the presence of a flame or heat source. **Flammability** is a critical property to consider for safety and regulatory purposes, influencing their use in construction, manufacturing, and transportation.

FLAMMABILITY

Example: Cotton fabric is flammable and can catch fire easily when exposed to a flame.

Specific Example: Polyethylene, a type of plastic, exhibits high flammability and can ignite quickly when exposed to a spark or flame.

REACTIVITY

Materials can react with other substances, leading to chemical changes. For instance, metals can react with acids to produce salts and hydrogen gas. Chemical reactivity varies widely among materials based on their composition and structure.

REACTIVITY

Example: Metals exhibit reactivity when they react with acids to produce salts and release hydrogen gas.

REACTIVITY

Specific Example: Iron undergoes a chemical reaction with oxygen and moisture in the air to form rust (iron oxide), which weakens the metal over time.

DEGRADABILITY

This refers to the ability of materials to break down physically or chemically over time. Degradable materials can be broken down into smaller components through processes such as weathering, mechanical action, or microbial activity.

DEGRADABILITY

Example: Leaves on the ground degrade naturally through exposure to weather and microbial activity.

Specific Example: Biodegradable plastics degrade into simpler substances when exposed to sunlight and water, unlike conventional plastics that persist for much longer.

BIODEGRADABILITY

Specifically, biodegradable materials can be broken down by biological organisms, such as bacteria and fungi, into simpler compounds like water, carbon dioxide, and biomass. Examples include natural fibers, certain plastics, and organic matter.

BIODEGRADABILITY

Example: Organic matter like food scraps can biodegrade when exposed to microbial activity in compost bins.

Specific Example: Biodegradable packaging made from cornstarch can be composted, breaking down into natural components without leaving harmful residues in the environment.

ASSESSMENT

GROUP ACTIVITY

1. The teacher will divide the class into five groups.
2. On half of a large sheet of cartolina each group will create a chart displaying examples of biodegradable and non-biodegradable materials.

ASSESSMENT

3. Each group will have the freedom to design their chart as they prefer.

4. Groups may use the provided chart below for reference.

ASSESSMENT

MATERIALS

NON-BIODEGRADABLE	BIODEGRADABLE

EXTENDED

Directions: Assess the learners using the rubric below.

EXTENDED

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Improvement (1)
Content Accuracy	Accurately identifies and categorizes examples of biodegradable and non-biodegradable materials.	Mostly identifies and categorizes examples correctly.	Some inaccuracies in identifying or categorizing materials.	Many inaccuracies in identifying or categorizing materials.
Clarity of Presentation	Information is clearly organized and easy to understand.	Information is organized and mostly clear.	Some sections lack clarity or organization.	Information is disorganized and difficult to understand.
Creativity and Design	Chart is highly creative, visually appealing, and effectively communicates the differences between biodegradable and non-biodegradable materials.	Chart is creative and visually appealing, effectively communicates differences.	Design is basic with limited creativity.	Little to no effort in design or creativity.
Use of Chart Space	Effectively uses space on cartolina with appropriate layout and utilization of visual elements.	Mostly uses space well with some overcrowding or underutilization.	Poor use of space, overcrowded or sparse presentation.	Chart lacks organization and effective use of space.
Adherence to Instructions	Fully adheres to all aspects of the given instructions.	Mostly follows instructions with minor deviations.	Significant deviations from instructions.	Does not follow instructions.